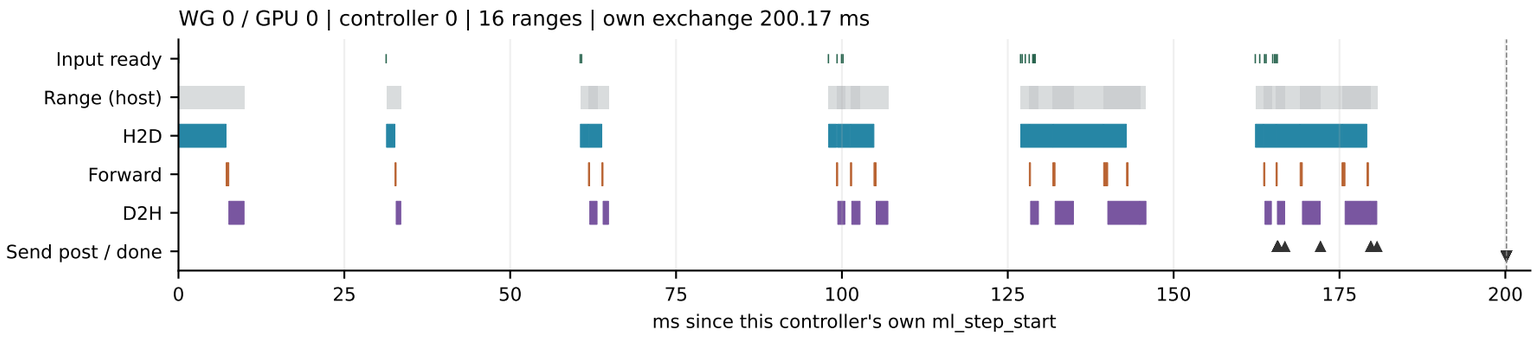
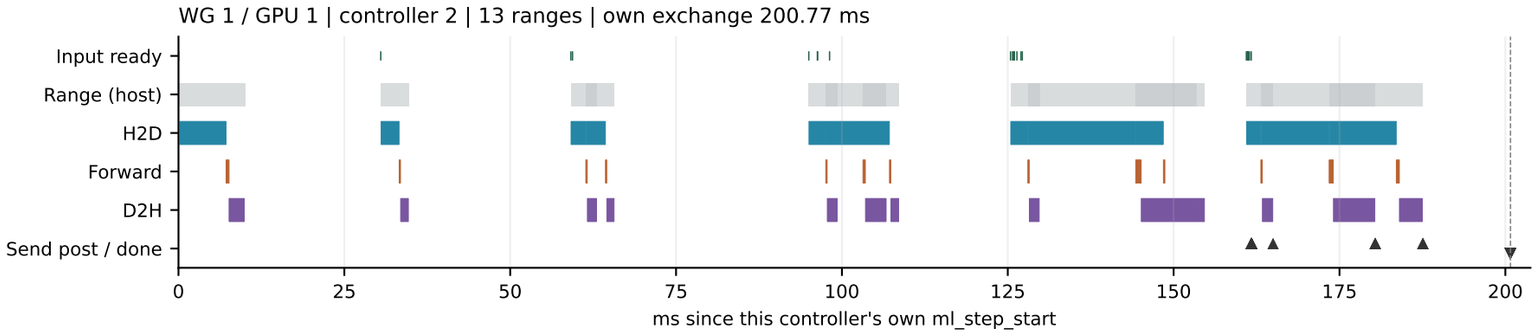
mlp | 96c4g | numa | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 388.14 ms | p2p1 341.94 ms



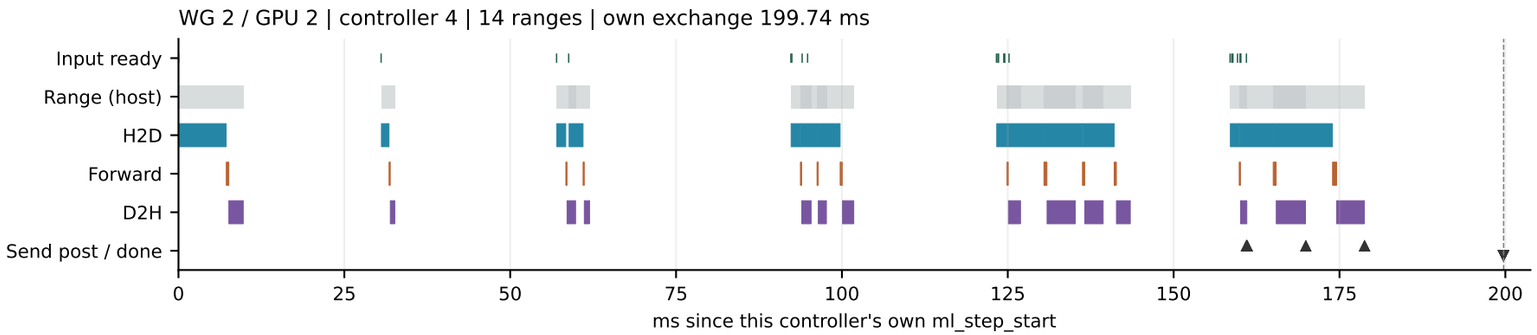
GPU work (ms)	Collective	p2p1
H2D	27.846	50.607
Forward	2.890	4.406
D2H	15.810	28.239
GPU SUM	46.546	83.251

Controllers: coll 0 / P2P 0
Coll host device elapsed: 46.745 ms
P2P GPU envelope: 180.526 ms
Different scopes; not an elapsed-time comparison.



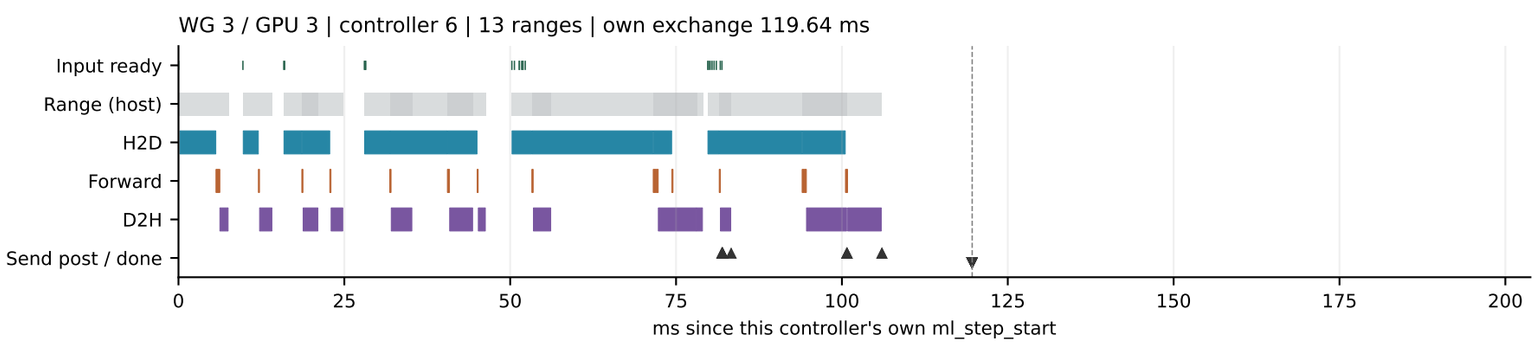
GPU work (ms)	Collective	p2p1
H2D	29.922	72.075
Forward	2.904	4.358
D2H	20.263	33.895
GPU SUM	53.090	110.328

Controllers: coll 2 / P2P 2
Coll host device elapsed: 53.355 ms
P2P GPU envelope: 187.284 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	29.335	51.837
Forward	2.896	4.188
D2H	14.844	29.739
GPU SUM	47.075	85.764

Controllers: coll 4 / P2P 4
Coll host device elapsed: 47.282 ms
P2P GPU envelope: 178.617 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	32.147	75.919
Forward	2.724	4.056
D2H	13.972	36.607
GPU SUM	48.843	116.582

Controllers: coll 6 / P2P 6
Coll host device elapsed: 49.041 ms
P2P GPU envelope: 105.717 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.